

NAME: _____

PERIOD: _____

One Result, Two Claims

AP Statistics · Unit 3 · Topic 3.13

[STAR] THE PROBLEM

Suppose a utility has 500,000 online accounts. It randomly selects 10,000, randomly assigns 5,000 to message A and 5,000 to B, and records on-time payment. The pre-data question asks whether the population proportions differ at $\alpha = 0.05$; deployment requires A to improve payment by at least 5 percentage points.

Rowan: “I rejected equality, so A has proved an important improvement of at least 5 points. Deploy A.”

Salma: “Rejecting equality supports a nonzero difference, but does not automatically establish the separate 5-point requirement.”

Conditions have been verified: random selection and assignment; $10,000 \leq 50,000$; pooled expected counts 3,000 and 2,000 in each group.

On-time payment

Text	n	Paid	Other
A	5000	3050	1950
B	5000	2950	2050

FIRST TAKE — one sentence before you work the parts

The observed improvement is 2 percentage points and the deployment target is 5. Is that enough to deploy? Give one reason.

[BOOK] CALCULATE, INTERPRET, THEN BOUND THE CLAIM (keep on the page)

For $H_0 : p_1 - p_2 = 0$, pool $\hat{p}_c = (x_1 + x_2)/(n_1 + n_2)$ and use $z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_c(1 - \hat{p}_c)(1/n_1 + 1/n_2)}}$. A two-sided test uses $P(|Z| \geq |z|)$. The p -value assumes H_0 is true and measures how often random chance would produce a difference at least as extreme. If the p -value is at most α , reject H_0 ; otherwise fail to reject it. A test against zero addresses whether a difference exists; it does not by itself establish that the difference exceeds a separate practical threshold. Printed normal table: $P(Z \leq 2.0412) = 0.979387$. A right-tail area is 1 minus the left area; the two-sided area is twice that tail.

Work (a)–(c).

DOK 1 (a) Calculate $\hat{p}_A$, $\hat{p}_B$, and the observed difference $\hat{p}_A - \hat{p}_B$.

DOK 2 (b) Using the verified conditions, calculate the pooled proportion, pooled SE, z , and the two-sided p -value for $H_0 : p_A - p_B = 0$.

DOK 3 (c) In 3–4 sentences, choose a report using the p -value, the 2-point observed gap, and the 5-point target. Explain what the test against zero establishes and why that does or does not justify deployment.

Frame: The p -value supports _____, because _____.

Frame: The test compares the difference with _____, but deployment requires _____; therefore _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored **E / P / I**.